Probability – 1

Expectation, indicators and linearity

Homework

11 сентября 2026 года

Theoretical minimum. Linearity of expectation without independence; the indicator method; the tail formulas $\mathrm{E}X = \sum_{k\geqslant 1} \Pr(X \geqslant k)$ and $\mathrm{E}X = \int_0^\infty \Pr(X \geqslant t)\,dt$; the law of total expectation and the recurrences it produces; stopping times and Wald's identity.

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1. (ISTCiM 2022) Let $k < n$. Let A be the event that n tosses of a fair coin contain at least k consecutive heads, and B the event that they contain at least $k+1$ consecutive heads or at least $k+1$ consecutive tails. Which of the two events is more probable?

2. (ISTCiM 2023) In a volleyball match each rally is served by the team that won the previous point, and the serving team wins a point with probability p. A set ends when a team has at least 25 points and leads by at least 2. The score is $24:22$ in favour of the team that won the last point. Find the expected number of further rallies.

3. (Mexmat, 2014) A computer picks a number uniformly at random from 1 to N. Two players guess in turn; each guess is a number not guessed before, and the computer answers whether the hidden number is equal, larger or smaller. The winner is the one who names it. Find the probability that the first player wins if both play optimally.

4. (MIPT) Petya invents a non-negative random variable ξ with $\mathrm{E}\xi = 1$; then Vasya, knowing the distribution of ξ, invents his own non-negative η, independent of ξ, with $\mathrm{E}\eta = 1$. For which least p can Petya guarantee $\Pr(\eta \geqslant \xi) \leqslant p$ whatever Vasya does?

5. (Сибирская 2022) A basket holds n balls, one white and the rest black. At each step two balls are drawn at random; if they differ in colour, both are repainted in one colour chosen with equal probability, and if they agree nothing happens. Find the expected number of steps until all balls have the same colour.

6. (Putnam 2022) Let $X_1, X_2, \dots$ be independent uniform on $[0,1]$ and put $S = \sum_{i=1}^{k} \frac{X_i}{2^i}$, where k is the least index with $X_k < X_{k+1}$ (and $k = \infty$ if there is none). Find $\mathrm{E}S$.

7. Prove Wald's identity and use it, together with the fair-game martingale, to compute the expected number of tosses of a fair coin until a prescribed finite pattern appears; check the answer on the patterns HH and HT.

8. Prove that the expected length of the cycle containing 1 in a random permutation of $\{1, \dots, n\}$ equals $\frac{n+1}{2}$, and that the expected number of cycles is H_n; deduce the asymptotics of both.

9. (Proposed problems) A stick is broken at two points chosen independently and uniformly, giving parts of lengths r, s, t. Show that the probability that there is a triangle whose sides are tangent to three mutually externally tangent circles of radii r, s, t equals $\frac{5}{27}$.

10. Prove the coupon collector estimate: the expected time to see all n coupons equals nH_n, and after $m \geqslant n \ln n + cn$ draws the probability that some coupon is still missing is at most e^{-c}.